Fairness-Targeted Random Walk for Negative Influence Minimization

A Unified Framework for Directed and Undirected Social Networks

Proposed Research Method

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Abstract

Negative influence minimization aims to reduce the harm caused by misinformation spreading through a social network. We consider a competitive-diffusion setting in which misinformation seeds are fixed and a limited number of truth seeds must be selected to protect vulnerable nodes. To focus on the least-protected users, this report proposes Fairness-Targeted Random Walk (FTRW), a candidate-selection framework for both directed and undirected graphs. FTRW estimates save probabilities by Monte Carlo simulation of the original competitive-diffusion model, assigns higher priority to vulnerable nodes with low estimated save probability, uses a personalized random walk to generate candidate truth seeds, and evaluates candidates against a maximin fairness objective. No surrogate estimator is used. The directed and undirected versions share the same framework but differ in the transition graph and distance calculations. FTRW is a proposed research method; no approximation guarantee or empirical superiority is assumed.

1 Introduction and Motivation

Influence-minimization interventions often select truth spreaders to compete with a fixed misinformation cascade. A method that maximizes total truth influence may improve the average outcome while leaving some vulnerable nodes poorly protected. We therefore focus on a fairness-oriented objective that seeks to maximize the save probability of the least-protected vulnerable node.

The proposed FTRW framework uses a random walk to generate a manageable set of candidate truth seeds. The random-walk score is used for candidate generation, not as a replacement for the diffusion model. Every shortlisted candidate is evaluated using direct Monte Carlo simulations of the original competitive-diffusion process. Thus, the framework does not use surrogate estimation.

2 Problem Definition

Let $G = (\mathcal{V}, \mathcal{E})$ be a social network, which may be directed or undirected. Let $\mathcal{M} \subseteq \mathcal{V}$ be the fixed misinformation-seed set, $\mathcal{V}_{\text{vul}} \subseteq \mathcal{V}$ the vulnerable-node set, and $\mathcal{T} \subseteq \mathcal{V} \setminus \mathcal{M}$ the selected truth-seed set. The truth-seed budget is k.

For a truth-seed set $\mathcal{T}$, define the save probability

$$\text{sp}_{\mathcal{T}}(v) = \Pr(v \text{ is saved from misinformation} \mid G, \mathcal{M}, \mathcal{T}), \tag{1}$$

where the event "saved" must be defined according to the original competitive-diffusion model.

The maximin objective is

$$\max_{\substack{\mathcal{T} \subseteq \mathcal{V} \setminus \mathcal{M} \\ |\mathcal{T}| \leq k}} \min_{v \in \mathcal{V}_{\text{vul}}} \text{sp}_{\mathcal{T}}(v). \tag{2}$$

This objective targets the worst-protected vulnerable node rather than only the aggregate number of users reached by truth.

3 Graph-Type Handling

The overall algorithm is the same for both graph types, but the graph used by the random walk and the definition of distance must be consistent with the network model.

3.1 Directed networks

For a directed graph, an edge $(u, v) \in \mathcal{E}$ represents a possible influence path from u to v, subject to the chosen diffusion model. Define G^{rev} by reversing every edge:

$$(u, v) \in \mathcal{E} \quad \Longrightarrow \quad (v, u) \in \mathcal{E}^{\text{rev}}.$$

The reverse graph is used for target-oriented candidate generation: a walk personalized toward vulnerable targets can assign scores to nodes that lie upstream of those targets in the original graph. Distances from misinformation seeds are measured along directed paths in the original graph.

3.2 Undirected networks

For an undirected graph, each edge is an unordered pair $\{u, v\}$, and traversal is possible in either direction under the graph representation. Reversing edges does not change the graph, so the walk is performed on G itself. The walk is still personalized toward vulnerable nodes with high protection deficits; its scores spread from those target nodes through the undirected network.

Distances from misinformation seeds are computed in the original undirected graph. For an unweighted graph, these are shortest-path distances; for a weighted graph, the distance definition must be chosen to match the intended interpretation of edge weights. If edge weights represent transmission probabilities rather than lengths, ordinary shortest-path distance may not be a suitable risk measure and should be treated only as a heuristic.

3.3 Unified transition-graph notation

Define the graph used for candidate generation as

$$G_{\text{walk}} = \begin{cases} G^{\text{rev}}, & \text{if } G \text{ is directed,} \\ G, & \text{if } G \text{ is undirected.} \end{cases} \tag{3}$$

Let P_{walk} be a row-stochastic transition matrix on G_{walk}, with a specified dangling-node rule. The transition probabilities must be defined consistently with the graph representation and the diffusion assumptions. For a simple unweighted walk, probability may be distributed uniformly over a node's neighbors; weighted variants require an explicitly stated normalization.

4 Proposed Method: FTRW

FTRW repeats the following steps until k truth seeds have been selected:

1. Estimate current node-level save probabilities using direct Monte Carlo simulation.
2. Compute protection deficits for vulnerable nodes.
3. Convert deficits, optionally combined with misinformation proximity, into an adaptive jump distribution.

4. Run a personalized random walk on G_{walk}.

5. Shortlist high-scoring eligible candidates.

6. Evaluate shortlisted candidates using direct simulations of the original competitive-diffusion model and select the candidate with the largest estimated maximin gain.

4.1 Monte Carlo estimation

At iteration i, let the current truth-seed set be $\mathcal{T}_i$. Run R independent simulations of the original competitive-diffusion model. Define

$$X_r(v;\mathcal{T}_i) = \begin{cases} 1, & \text{if } v \text{ is saved in simulation } r, \\ 0, & \text{otherwise.} \end{cases}$$

Estimate the save probability by

$$\widehat{\text{sp}}_{\mathcal{T}_i}(v) = \frac{1}{R}\sum_{r=1}^{R} X_r(v;\mathcal{T}_i). \tag{4}$$

This is direct Monte Carlo estimation, not surrogate estimation. Use independent simulation trials for final method evaluation. Common random numbers may be used for candidate comparisons if supported consistently by the simulator, but final reported performance should be evaluated on fresh trials.

4.2 Protection deficit

Let $\theta \in [0,1]$ be a target save-probability threshold. Define

$$d_i(v) = \max\left\{0, \theta - \widehat{\text{sp}}_{\mathcal{T}_i}(v)\right\}. \tag{5}$$

A vulnerable node below the target receives positive deficit, with larger deficits assigned to less-protected nodes. The threshold is a tuning parameter and does not guarantee that every node can reach it.

4.3 Misinformation-proximity factor

Let $\text{dist}_G(\mathcal{M}, v)$ denote the shortest-path distance from any misinformation seed to v, using the original graph and respecting edge directions when G is directed. Define

$$e(v) = \begin{cases} \exp\left(-\dfrac{\text{dist}_G(\mathcal{M},v)}{\tau}\right), & \text{if } \text{dist}_G(\mathcal{M},v) < \infty, \\ 0, & \text{otherwise,} \end{cases} \tag{6}$$

where $\tau > 0$ controls decay. This is a heuristic proxy for exposure risk; it is not an exact probability of misinformation reaching a node. For directed graphs, a node unreachable from all misinformation seeds receives $e(v) = 0$. For undirected graphs, distance is symmetric.

4.4 Adaptive target weights

Let $\eta \in [0,1]$ control the relative contribution of deficit and proximity. Define

$$h_i(v) = \begin{cases} d_i(v)\left[\eta + (1-\eta)e(v)\right], & v \in \mathcal{V}_{\text{vul}}, \\ 0, & v \notin \mathcal{V}_{\text{vul}}. \end{cases} \tag{7}$$

When $\eta = 1$, only protection deficit affects the weights. Smaller values increase the relative influence of the proximity factor.

Define the personalized jump distribution using $\varepsilon \in (0, 1]$:

$$q_i(v) = \begin{cases} \dfrac{\varepsilon}{|\mathcal{V}|} + (1-\varepsilon)\dfrac{h_i(v)}{\sum_{u\in\mathcal{V}} h_i(u)}, & \displaystyle\sum_{u\in\mathcal{V}} h_i(u) > 0, \\ \dfrac{1}{|\mathcal{V}|}, & \displaystyle\sum_{u\in\mathcal{V}} h_i(u) = 0. \end{cases} \tag{8}$$

The uniform component supports exploration. If all deficits are zero, the jump distribution becomes uniform.

4.5 Personalized random walk

Let $\mathbf{q}_i$ be the column vector corresponding to q_i, and let $\boldsymbol{\pi}_i$ be the column vector of random-walk scores. Using the transition matrix P_{walk} on the graph specified in Equation (3), compute

$$\boldsymbol{\pi}_i = \alpha \mathbf{q}_i + (1-\alpha) P_{\text{walk}}^{\mathsf{T}} \boldsymbol{\pi}_i, \tag{9}$$

where $\alpha \in (0, 1]$ is the restart probability. Equivalently,

$$\boldsymbol{\pi}_i = \alpha \left[I - (1-\alpha) P_{\text{walk}}^{\mathsf{T}} \right]^{-1} \mathbf{q}_i, \tag{10}$$

when the inverse is defined. In practice, the fixed point is computed iteratively.

For a directed graph, P_{walk} is constructed from G^{rev}; for an undirected graph, it is constructed from G. For undirected graphs, the walk's edge transition probabilities should still be defined carefully, especially when degrees vary or the graph is weighted. The walk score is a candidate-generation heuristic, not an estimate of save probability.

4.6 Candidate shortlist

Let L be the shortlist size. Exclude nodes that are already misinformation seeds or selected truth seeds:

$$C_i = \text{TopL}_{v\in\mathcal{V}\setminus(\mathcal{M}\cup\mathcal{T}_i)}\, \pi_i(v), \tag{11}$$

where TopL returns up to L eligible nodes with the highest scores. Shortlisting reduces simulation cost, but may exclude the best candidate; its effect must be measured.

4.7 Direct candidate evaluation

Define the estimated minimum save probability under seed set $\mathcal{T}_i$:

$$\widehat{m}(\mathcal{T}_i) = \min_{v\in\mathcal{V}_{\text{vul}}} \widehat{\text{sp}}_{\mathcal{T}_i}(v). \tag{12}$$

For each $x \in C_i$, estimate save probabilities for $\mathcal{T}_i \cup \{x\}$ by direct Monte Carlo simulation. Define the estimated maximin gain:

$$\widehat{\Delta}_{\min}(x \mid \mathcal{T}_i) = \widehat{m}(\mathcal{T}_i \cup \{x\}) - \widehat{m}(\mathcal{T}_i). \tag{13}$$

Choose

$$x_i^\star \in \arg\max_{x\in C_i} \widehat{\Delta}_{\min}(x \mid \mathcal{T}_i), \qquad \mathcal{T}_{i+1} = \mathcal{T}_i \cup \{x_i^\star\}. \tag{14}$$

An optional tie-breaker uses the r worst-off vulnerable nodes under the current seed set. Let $W_i \subseteq \mathcal{V}_{\text{vul}}$ contain the r nodes with the lowest current estimated save probabilities. Among candidates with statistically indistinguishable maximin gains, prefer the candidate maximizing

$$\widehat{a}_r(\mathcal{T}_i \cup \{x\}) = \frac{1}{r} \sum_{v \in W_i} \widehat{\text{sp}}_{\mathcal{T}_i \cup \{x\}}(v). \tag{15}$$

Keep W_i fixed while comparing candidates in iteration i. Evaluate this tie-breaker as a separate ablation.

5 Algorithm

Algorithm 1 FTRW for Directed and Undirected Graphs

Require: Graph $G = (\mathcal{V}, \mathcal{E})$; graph type; misinformation seeds $\mathcal{M}$; vulnerable set $\mathcal{V}_{\text{vul}}$; budget k; parameters $\theta, \eta, \varepsilon, \alpha, \tau$; shortlist size L; simulation count R

Ensure: Truth-seed set $\mathcal{T}$, with $|\mathcal{T}| \leq k$

if G is directed **then**
  $G_{\text{walk}} \leftarrow G^{\text{rev}}$
  Compute directed distances from $\mathcal{M}$ in original graph G
else
  $G_{\text{walk}} \leftarrow G$
  Compute undirected distances from $\mathcal{M}$ in G
end if
$\mathcal{T} \leftarrow \emptyset$
for $i = 0, \ldots, k-1$ **do**
  Estimate $\widehat{\text{sp}}_{\mathcal{T}}(v)$ for $v \in \mathcal{V}_{\text{vul}}$ using R original-model simulations
  Compute $d_i(v)$, $e(v)$, $h_i(v)$, and $q_i(v)$
  Compute $\boldsymbol{\pi}_i$ on G_{walk} using Equation (9)
  Form C_i using Equation (11)
  for all $x \in C_i$ **do**
    Estimate $\widehat{m}(\mathcal{T} \cup \{x\})$ by direct Monte Carlo simulation
    Compute $\widehat{\Delta}_{\min}(x \mid \mathcal{T})$
  end for
  Select $x_i^\star$ using Equation (14)
  $\mathcal{T} \leftarrow \mathcal{T} \cup \{x_i^\star\}$
end for
return $\mathcal{T}$

6 Experimental Design

6.1 Baselines

Compare FTRW against:

- **PROTECT**, reproduced from its original specification.
- **Random selection** of truth seeds.
- **An aggregate-influence baseline**, if compatible with the same diffusion model.
- **FTRW ablations**: uniform restart, deficit-only weighting, proximity-aware weighting, and alternative random-walk settings.

- **Exhaustive greedy maximin selection** on small graphs, evaluating every eligible candidate by direct Monte Carlo simulation when feasible.

Use the same graph, misinformation seeds, vulnerable set, budget, diffusion assumptions, and evaluation protocol for all comparable methods.

6.2 Metrics

The primary metric is the minimum vulnerable-node save probability:

$$\widehat{M}_{\min}(\mathcal{T}) = \min_{v \in \mathcal{V}_{\text{vul}}} \widehat{\text{sp}}_{\mathcal{T}}(v). \tag{16}$$

Also report mean save probability and the 10th percentile:

$$\widehat{M}_{\text{mean}}(\mathcal{T}) = \frac{1}{|\mathcal{V}_{\text{vul}}|} \sum_{v \in \mathcal{V}_{\text{vul}}} \widehat{\text{sp}}_{\mathcal{T}}(v), \tag{17}$$

$$\widehat{Q}_{0.10}(\mathcal{T}) = \text{Quantile}_{0.10}\left(\{\widehat{\text{sp}}_{\mathcal{T}}(v) : v \in \mathcal{V}_{\text{vul}}\}\right). \tag{18}$$

Report runtime, number of simulations, and results across different truth-seed budgets k.

6.3 Experiments

1. **Graph-type comparison:** evaluate directed and undirected networks separately; do not assume results transfer between them.
2. **Budget sensitivity:** vary k and measure minimum and mean save probability.
3. **Parameter sensitivity:** vary $\theta, \eta, \varepsilon, \alpha, \tau$, and L.
4. **Ablation study:** isolate the effects of deficit weights, proximity, transition graph, and tie-breaking.
5. **Shortlist quality:** compare shortlisting with exhaustive candidate evaluation on small instances.
6. **Runtime–quality trade-off:** plot protection metrics against wall-clock time and simulation count.
7. **Robustness:** repeat experiments across multiple misinformation seed sets, vulnerable sets, and random seeds.

Use independent Monte Carlo trials for final evaluation and report confidence intervals or variability across repeated runs. When appropriate, use paired randomness for comparing methods, and avoid claiming improvements smaller than the uncertainty in the estimates.

7 Computational Complexity

Let $n = |\mathcal{V}|$, $m = |\mathcal{E}|$, J be the number of random-walk iterations, L the shortlist size, R the simulations per candidate, and C_{sim} the average cost of one diffusion simulation. For unweighted graphs, computing distances from all misinformation seeds costs $O(n + m)$ using multi-source breadth-first search. A sparse random-walk iteration costs $O(n + m)$, so recomputing scores for k rounds costs approximately

$$O(kJ(n + m)).$$

A straightforward candidate-evaluation stage costs approximately

$$O(kLRC_{\text{sim}}),$$

in addition to the baseline save-probability estimates. These are high-level estimates; actual costs depend on diffusion dynamics, caching, parallelism, and graph representation. Direct simulation may dominate runtime. No surrogate estimator is introduced.

8 Limitations and Research Risks

- **No approximation guarantee:** no approximation ratio is claimed for the maximin objective or FTRW.
- **Heuristic random-walk ranking:** a high walk score does not guarantee a high marginal improvement in save probability.
- **Proximity approximation:** shortest-path distance is not a complete model of stochastic misinformation exposure.
- **Monte Carlo noise:** candidate rankings may change due to simulation variance.
- **Shortlist error:** the best candidate under direct evaluation may not appear in the shortlist.
- **Graph-model dependence:** directed and undirected networks encode different reachability assumptions; transition probabilities must match the intended diffusion model.
- **Vulnerable-set dependence:** fairness results depend on how the vulnerable population is defined.
- **Novelty is unverified:** the combination of adaptive target weighting, personalized random walks, and direct maximin candidate evaluation must be compared with prior work before claiming originality.

9 Recommended Development Plan

1. Reproduce the original competitive-diffusion model, save event, and PROTECT baseline.
2. Validate direct Monte Carlo estimates on small graphs.
3. Implement FTRW for directed graphs using the reversed graph for candidate generation.
4. Implement FTRW for undirected graphs using the original graph.
5. Start with deficit-only weights, then add the proximity factor.
6. Compare shortlist-based selection with exhaustive candidate evaluation on small instances.
7. Add LFPR-/FSPR-inspired variants only after reproducing the exact source algorithms.
8. Run repeated trials, uncertainty analysis, parameter sensitivity, and ablations separately for each graph type.
9. Conduct a focused literature review before claiming novelty or superiority.

10 Conclusion

This report proposes a unified FTRW framework for negative influence minimization on directed and undirected social networks. Both versions prioritize vulnerable nodes with low estimated save probability, use personalized random walks to generate candidate truth seeds, and evaluate candidates by direct Monte Carlo simulation of the original competitive-diffusion model. For directed networks, the random walk uses the reversed graph to find potential upstream truth seeds; for undirected networks, it uses the original graph because edge reversal leaves the graph unchanged.

The central research question is:

> Can fairness-targeted personalized random walks generate truth-seed candidates that improve the minimum save probability of vulnerable nodes more efficiently than PROTECT on both directed and undirected networks?

This is an empirical question. The framework should be treated as a research proposal, not a proven improvement. The save-probability definition and the exact PROTECT, LFPR, and FSPR specifications must be checked against their original sources before implementation and publication.